

VANCE AFB, OK, USA

WMO: 723535

Lat: 36.333N

Lon: 97.917W

Elev: 398

StdP: 96.63

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-9.1	-17.5	0.8	-9.9	-14.7	1.1	-6.4	14.9	3.6	12.9	6.3	4.9	10	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	38.7	23.0	37.3	23.1	35.7	23.4	25.7	32.4	25.1	32.1	24.6	31.5	6.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	20.0	27.6	23.4	19.1	27.3	22.7	18.3	27.1	81.5	32.6	78.7	32.0	76.4	31.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.6	10.5	-15.3	41.2	2.7	1.4	-17.3	42.2	-18.8	43.0	-20.3	43.8	-22.3	44.8		
(5)			-15.5	26.8	2.6	0.8	-17.3	27.4	-18.8	27.9	-20.3	28.4	-22.1	29.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	2.2	5.4	9.9	14.1	20.2	25.9	28.3	27.5	23.1	16.1	9.1	3.0		
	DBStd	10.35	5.70	6.58	6.07	4.96	4.33	3.33	2.82	3.03	4.20	5.07	5.15	4.88		
	HDD10.0	808	248	154	78	18	0	0	0	0	11	77	223			
	HDD18.3	2171	501	362	267	142	31	1	0	0	11	102	278	476		
	CDD10.0	2798	5	26	73	141	317	476	568	544	393	199	49	6		
	CDD18.3	1121	0	1	5	15	89	227	309	286	154	33	2	0		
	CDH23.3	12745	1	12	61	198	881	2572	3864	3355	1468	317	18	0		
	CDH26.7	6487	0	2	14	45	333	1329	2187	1828	654	92	2	0		
(14) Wind	WSAvg	5.3	5.2	5.4	6.0	6.1	5.2	5.5	4.8	4.6	4.7	5.3	5.3	5.1		
(15) Precipitation	PrecAvg	815	25	31	58	84	98	119	77	91	65	68	50	38		
	PrecMax	1258	68	103	181	195	252	331	152	226	134	187	145	97		
	PrecMin	504	0	0	1	11	11	17	2	6	0	2	1	3		
	PrecStd	180	20	28	44	46	60	73	36	54	32	49	41	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	24.8	28.8	30.4	35.1	38.8	40.9	40.7	37.5	32.1	25.9	19.6		
		MCWB	11.5	14.9	16.5	18.2	22.4	23.5	22.8	22.8	22.0	21.1	16.9	12.0		
	2%	DB	17.2	21.7	24.8	27.9	32.4	36.7	38.9	38.5	34.6	29.2	22.6	16.8		
		MCWB	9.8	12.5	16.0	17.6	22.1	23.3	23.0	23.0	23.1	19.9	15.0	11.3		
	5%	DB	14.1	18.9	22.2	25.7	30.4	35.1	37.4	37.1	32.7	27.0	20.1	13.4		
		MCWB	8.7	12.2	14.8	16.8	21.4	23.4	23.4	22.8	22.7	18.5	14.0	8.9		
	10%	DB	11.3	15.9	19.6	23.3	28.2	33.4	35.7	35.1	30.7	24.7	17.5	11.3		
		MCWB	6.8	10.5	13.4	15.8	20.6	23.0	23.7	23.3	22.1	17.6	12.6	7.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	17.2	19.1	21.1	25.0	26.4	26.7	26.5	25.6	23.7	18.6	16.9		
		MCDB	17.4	19.6	24.3	26.5	29.9	33.4	32.5	32.8	30.6	28.3	22.5	18.4		
	2%	WB	11.2	15.0	17.7	19.9	23.8	25.4	25.6	25.4	24.7	22.2	16.7	12.1		
		MCDB	15.0	19.5	22.1	24.9	29.2	33.0	32.5	32.2	30.2	26.0	20.3	14.2		
	5%	WB	8.9	13.3	16.3	18.8	22.8	24.7	25.0	24.8	23.8	20.7	15.3	9.5		
		MCDB	13.6	17.3	19.9	23.1	28.3	32.0	32.3	31.7	29.3	24.1	18.6	13.2		
	10%	WB	7.0	10.8	14.6	17.2	21.8	23.9	24.4	24.1	23.0	19.0	13.6	7.3		
		MCDB	10.9	16.5	18.9	21.7	26.9	30.9	31.9	31.2	28.4	23.3	17.1	10.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.9	12.9	13.2	12.1	12.3	12.5	12.2	11.6	12.3	12.5	11.3		
		MCDBR	17.1	17.3	16.6	16.6	14.4	14.2	14.9	15.1	13.7	14.8	15.5	15.6		
		MCWBR	10.4	10.2	9.4	8.8	6.2	4.5	3.8	4.1	4.7	7.0	9.2	10.5		
		MCDBR	15.2	14.4	13.6	13.3	12.5	12.7	12.5	11.8	11.7	11.1	12.9	13.9		
	5% WB	MCWBR	10.2	9.8	9.0	8.4	6.5	5.2	4.2	4.3	4.6	6.8	9.1	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.294	0.335	0.373	0.406	0.417	0.420	0.424	0.380	0.343	0.301	0.282		
	taud		2.501	2.496	2.402	2.319	2.304	2.311	2.329	2.265	2.367	2.456	2.507	2.522		
	Ebn at Noon		918	952	933	909	879	865	860	850	873	880	891	901		
	Edh at Noon		80	91	110	126	130	130	126	132	113	94	80	74		
(44) All-Sky Solar Radiation	RadAvg		2.73	3.40	4.45	5.53	6.12	6.91	6.86	6.18	5.15	3.87	2.91	2.32		
	RadStd		0.27	0.49	0.31	0.39	0.61	0.55	0.38	0.35	0.38	0.48	0.36	0.23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page